

# KOUSHOR ROY CHOUDHURY

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## EDUCATION

### Institute of Engineering and Management, Kolkata

Aug 2023 – May 2027

B.Tech in Information Technology — CGPA: 9.44 / 10

Kolkata, India

**Relevant Coursework:** Data Structures & Algorithms, Operating Systems, Object-Oriented Programming, Database Management Systems, Computer Networks, Software Engineering

## WORK EXPERIENCE

### 3W Business Private Limited

July 2026 – Present

**Full Stack Developer Intern** · React.js, Node.js, Express.js, MongoDB

Remote

- Designed and implemented an optimized real-time user status tracking system using Socket.io leveraging efficient state management and backend optimization techniques, reducing server and database load by **30%** while ensuring accurate live status updates.
- Optimized the Points History module by redesigning the MongoDB aggregation pipelines and data retrieval logic, reducing page load time by **50%** while validating performance on datasets containing **50,000+ records**.
- Engineered a configurable URL template management system supporting **3 customizable link templates**, replacing random URL generation and improving consistency, maintainability, of shared links.

## PROJECTS

### AuraSync: AI Emotion-Based Music Recommendation System

Sept 2025 – Nov 2025

React.js, Node.js, Spotify API, face-api.js | [GitHub](#)

- Applied a machine learning model capable of detecting **5 facial emotions** from real-time user images to drive music recommendations.
- Developed a system that dynamically recommends songs based on detected emotions and updates the website theme in under **2 seconds**.
- Integrated Spotify OAuth to let users connect accounts and receive personalized, emotion-aware music recommendations.
- Optimized facial feature processing from real-time user images, improving recommendation accuracy and reducing site load time by **50%**.

### Smart Leads Dashboard

Jan 2026 – Feb 2026

React.js, Node.js, Express.js, MongoDB, RBAC | [GitHub](#)

- Built a full-stack lead management dashboard enabling users to create, update, filter, and track **500+ customer leads** through an intuitive interface.
- Implemented Role-Based Access Control (RBAC) with Admin and User roles, ensuring secure access to lead management features and sensitive customer data.
- Built secure RESTful APIs using Express.js and Node.js to manage lead data, improving data retrieval and update efficiency.

### Library Management System

Mar 2025 – Apr 2025

Java, OOP, Collections Framework

- Developed a Java-based library management system supporting adding, searching, updating, borrowing, returning, reserving, and sorting books, reducing check-out times by **25%**.
- Implemented an overdue-tracking system to surface all overdue books and calculate fines automatically.
- Managed library data for **100+ books** with efficient search and retrieval mechanisms.

## TECHNICAL SKILLS

- Programming Languages:** C, Python, Java, JavaScript, TypeScript
- Web Technologies:** HTML, CSS, React.js, Node.js, Express.js, REST APIs, Next.js
- Databases:** MongoDB, MySQL, Supabase, Redis
- Development Tools:** Git, GitHub, Postman, Docker
- Competitive Programming:** Solved 500+ problems across CodeChef, Codeforces, and LeetCode

## ACHIEVEMENTS

- Google Cloud Computing Foundation – NPTEL (IIT Kharagpur), 2024:** Covered cloud architecture, virtualization, storage, networking, and application deployment through practical labs and real-world use cases.
- Academic Excellence Award, IEM:** Ranked 1st in the IT department in 1st year and 4<sup>th</sup> in 2<sup>nd</sup> year.
- Lead, Co-Lead, IEM Gen-AI COE,** leading the organization of Gen-AI workshops, technical events, and student engagement initiatives.